

Spline Regression

2026-04-17

Introduction

Splines are piecewise polynomials joined smoothly at points called knots. They give flexible non-linear fits without the wild extrapolation behaviour of high-degree polynomials, which makes them the canonical tool for modelling non-linear continuous predictor effects in regression. Natural cubic splines and restricted cubic splines are the workhorses in clinical and epidemiological work; their constraint of linear behaviour beyond the boundary knots prevents the implausible curls that ordinary polynomials produce in the tails.

Prerequisites

A working understanding of polynomial regression, the limitations of high-degree polynomials at the extremes of the data, and the concept of basis functions in regression.

Theory

A natural cubic spline with k internal knots is a piecewise cubic polynomial that is continuous together with its first and second derivatives at each knot, and linear beyond the boundary knots. It uses $k + 1$ degrees of freedom for the non-linear effect. Knot placement is typically at quantiles of X (3 knots: 10/50/90; 4 knots: 5/35/65/95; 5 knots: 5/27.5/50/72.5/95 percentiles). Three to five knots are adequate for most biomedical applications; more invites over-fitting.

Restricted cubic splines (RCS) impose linearity beyond the outer knots, preventing the polynomial from curling away in the tails — useful for clinical reporting where extrapolation into low-data regions is a real concern.

Assumptions

Standard regression assumptions on residuals; residuals vs. fitted should show no remaining structure after the spline is included; the chosen number and placement of knots are adequate for the underlying functional form.

R Implementation

```
library(splines); library(rms)

set.seed(2026)
x <- runif(200, 0, 10)
y <- 2 + sin(x) * 2 + rnorm(200, 0, 0.5)

# Natural cubic spline via splines::ns
fit_ns <- lm(y ~ ns(x, df = 4))
summary(fit_ns)

# Visualise
plot(x, y, pch = 20)
xg <- seq(0, 10, length.out = 200)
```

```
lines(xg, predict(fit_ns, newdata = data.frame(x = xg)),
      col = "#2A9D8F", lwd = 2)

# rms::rcs for restricted cubic splines with plot
dd <- datadist(data.frame(x, y)); options(datadist = "dd")
fit_rcs <- ols(y ~ rcs(x, 4))
plot(Predict(fit_rcs, x = seq(0, 10, 0.2)))
```

Output & Results

`splines::ns()` and `rms::rcs()` produce basis matrices that enter `lm()` like any other predictor. The resulting fit object has the usual summary, coefficients (which are not individually interpretable), and predictions. Visualising the partial effect of X on Y via `predict()` or `rms::Predict()` is the primary inferential output.

Interpretation

A reporting sentence: “A natural cubic spline with 4 degrees of freedom captured the underlying sinusoidal pattern of y on x ($R^2 = 0.92$); a likelihood-ratio test against the linear-only model strongly rejected linearity ($\chi^2_3 = 287$, $p < 0.001$).” Always test for non-linearity formally and report the test result; a smooth curve in a plot is not enough.

Practical Tips

- Use 3–5 knots for most applications; more risks overfitting and hard-to-interpret partial effects.
- Restricted cubic splines (`rms::rcs`) constrain the tails to be linear, avoiding the wild extrapolation that natural cubic splines can produce when knots are sparse.
- Test for non-linearity with a likelihood-ratio test of the spline model against a linear-only specification; report the test result alongside any non-linear visualisation.
- Spline coefficients are not individually interpretable; communicate results via `Predict()` plots or `ggeffects::ggpredict()`.
- For multiple continuous predictors with potential non-linear effects, consider GAMs (`mgcv::gam`) which automate smoothness selection.
- For survival or logistic regression, the same `rcs()` syntax inside `cph()` or `lrm()` extends the spline machinery to the relevant likelihood.

R Packages Used

`splines` for `ns()` and `bs()` basis functions; `rms` for `rcs()`, `Predict()`, and `datadist`-aware regression workflows; `mgcv` for GAMs as the more general alternative; `ggeffects` for predicted-effect visualisation; `Hmisc::rcspline.eval()` for low-level RCS basis construction.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression

- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI